

PRASAD MK

Solution Architect · Agentic AI & Backend · SRE, Infrastructure & Platform Engineer

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SUMMARY

Technologist with 25+ years designing, operating, and scaling large-scale distributed systems, including 8 years as sole SRE owner of a multitenant AIOps SaaS platform holding 99.99% uptime across the Americas. Owned full-lifecycle production readiness, capacity planning, and on-call incident response for a platform processing 8TB/day at sub-second latency, cutting incident resolution time 45% and replacing manual toil with automation to save \$2.1M a year in infrastructure cost. Directly supported Apple's own tenancy on this platform, keeping Operations Agent and Performance Agent telemetry flowing through Cloud Optimizer into Apple's Vertica data farm. Now building agentic AI backend services on AWS Bedrock with multi-agent orchestration and RAG pipelines, and driving AI-assisted tooling adoption into day-to-day reliability workflows. Published researcher on AI agent security policy and trust-boundary frameworks, with a pending patent on Generation-Bound Cache Invalidation.

TECHNICAL SKILLS

Languages	Python, Go, Java (Spring Boot), Bash, Perl, C/C++, SQL, Javascript
AI/Agentic Systems	AWS Bedrock multi-agent orchestration, RAG pipelines, vector retrieval, AWS SageMaker, ML-based event correlation, model deployment automation, AI-assisted diagnostics and copilot tooling adoption
Reliability Practice	SLO/SLI ownership, production readiness reviews, capacity planning, on-call incident response, runbook-as-code, blameless postmortems
GCP	GKE, Cloud Monitoring, Vertex AI, Cloud Functions, cross-cloud metric collection and normalization
AWS	EKS, EC2, VPC, S3, RDS, Lambda, Bedrock, Cost Explorer, SageMaker, CloudWatch
IaC	Terraform, Ansible, CloudFormation; multi-cloud architecture (AWS + GCP); bare-metal and hybrid topology design
Governance & Compliance	Agent trust boundary design, SOC2 compliance, DR strategy, Access Controls, Security Scanning (0% critical vulns in prod)
Containers & CI/CD	Kubernetes, OpenShift, Helm, Docker, ArgoCD, Jenkins, GitLab CI, GitHub Actions
Observability	Dynatrace, Prometheus, Grafana, OpenTelemetry, HPE/Micro Focus AIOps stack (Operations Agent, Performance Agent, OMi, Cloud Optimizer)
Data & Streaming	Kafka, Pulsar, Kafka Connect (8TB/day at sub-second latency); PostgreSQL, MS SQL, DynamoDB, Vertica
Systems	Linux internals (RHEL, Ubuntu, SUSE), TCP/IP, DNS, TLS, IPv4/IPv6 migration, packet-level tracing

PROFESSIONAL EXPERIENCE

Solutions Architect | Technical Lead

Wipro Ltd. (Client: PNC Bank)

Oct 2025 – Present

Pittsburgh, PA

- Own architecture and production readiness for a microservices-based Alert Management System on OpenShift/Kubernetes with a full Dynatrace observability stack, setting SLO targets and automated alerting that catches degradation in under 5 minutes across 70,000 users.
- Lead capacity planning and Platform-as-Code provisioning standards on Bitbucket, enabling on-call engineers to diagnose and resolve incidents without escalation.
- Write backend microservices in Java/Spring Boot with JUnit/Mockito coverage on critical paths; architect CI/CD pipelines in Jenkins and Terraform with embedded security scanning, holding production at 0% critical vulnerabilities.
- Partner across development and operations teams to drive continuous deployment adoption; lead rollout of Copilot-based AI tooling across engineering workflows, cutting time spent on boilerplate and code review.
- Implement Kubernetes Vertical Pod Autoscaler across the organization for better resource planning and allocation.
- Building a Kafka-based branch closure notification system, publishing closure events to downstream consumers in real time so affected teams and customer-facing systems could react without polling or manual coordination.

Software Engineer, Pre-Sales AI Systems

Kakudi

Jun 2025 – Sep 2025

Folsom, CA

- Built agentic AI systems on AWS Bedrock with multi-agent orchestration, RAG pipelines, and automated CI/CD model deployment; built Voice AI agents pulling from live data sources.
- Designed and deployed cloud-native microservices on Kubernetes using Terraform-provisioned AWS infrastructure; wrote Python automation and Ansible playbooks for operational workflows.
- Contributed to proposals and statements of work (SOWs) and presented technical solutions directly to prospective clients.
- Ran client discovery calls and scoped project costs and timelines for client engagements, mainly non-profit organizations.

HP, HPE, Micro Focus, and OpenText are one continuous employer, joined by a corporate split and two acquisitions. The four roles below span 21 years in the same systems management product line, from on-premise agents to today's AIOps SaaS platform.

Site Reliability Engineer

Micro Focus / OpenText

Sep 2017 – May 2025

Rocklin, CA

- Progressed from DevOps Engineer to full-stack SRE owner of Operations Bridge Manager, an enterprise AIOps SaaS platform serving customers across the Americas; held 99.99% SLA uptime for 8 straight years and cut incident resolution time 45% through Linux kernel diagnostics and packet-level tracing.
- Ran production readiness reviews and capacity planning for Apple's own tenancy on the multi-tenant OMi platform, keeping Operations Agent and Performance Agent telemetry flowing through Cloud Optimizer into Apple's Vertica data farm.
- Built the Automated Event Correlation (AEC) engine, a machine learning model that groups related alerts by topology and event history, cutting alert noise and surfacing root-cause events faster during high-volume incidents.
- Developed the Agent Metric Collector, a Go-based service pulling and normalizing performance metrics from AWS and GCP into a central correlation engine for cross-cloud alerting.
- Led migration of 50+ microservices to Kubernetes/OpenShift, cutting multi-tenant provisioning time from days to 30 minutes and saving \$2.1M a year in cloud infrastructure cost through right-sizing and Terraform automation.
- Built Kafka and Pulsar telemetry pipelines processing 8TB/day at sub-second latency; built a Python-based ML cost optimizer against the AWS Cost Explorer API that cut quarterly spend 20%.
- Owned SOC2 compliance and DR strategy for the organization, including quarterly audits of access controls and security policy.

Product Engineer | Software Designer

Hewlett Packard Enterprise (HPE)

Nov 2015 – Aug 2017

Sunnyvale, CA

- Sustained Fortune 500 technical escalation leadership through the HP/HPE corporate split, holding a 100% resolution rate on high-priority incidents.
- Built a Patch Manager that automated hotfix packaging, validation, and rollout for Operations Agent, cutting manual patch errors and customer-impacting regressions.
- Designed KPI collection frameworks for KVM and Helium hypervisors, feeding metrics into centralized observability stacks and improving visibility across multi-tenant environments.
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Product Engineer | Software Designer

Hewlett Packard (HP)

Sep 2004 – Oct 2015

Sunnyvale, CA

- Scaled the Backbone Communication Component from 2,000 to 10,000 concurrent connections and built a reverse channel proxy that let traffic through a single outbound connection with no inbound firewall rules.
- Led the IPv4-to-IPv6 migration of the Operations Agent network stack with zero disruption; wrote embedded Perl memory caching that let the agent run two years straight in mission-critical environments.
- Built a custom log hashing algorithm to catch meaningful changes in log files, cutting false-positive alerts by over 60% in large environments.
- Built a centralized install repository for Operations Agent for remote, repeatable deployments at scale, cutting manual provisioning effort by 80%.
- Resolved 100% of high-priority on-site escalations at Apple, Samsung, Disney, FedEx, and PG&E through root-cause analysis at the architecture level. Won HP's Global High Impact Award.

Software Engineer

ProcSys (Client: Cisco)

Sep 1999 – Aug 2004

Bangalore, India

- Built OpsXML, an intent-driven, declarative framework for automating Cisco router configuration — one of the earliest infrastructure-as-code implementations for physical networking.
- Built an eLearning platform for Cisco devices used by thousands of students preparing for the CCNA exam.

PUBLICATIONS & ADDITIONAL INFO

- "Agent Security Policy (ASP)" and "Escalation Threshold Theory (SET) Pre-prints," SSRN Author ID 10270516.
- Day One AI: From Your First Job to Your First Product, published book, Amazon Kindle (ISBN 979-8-9957306-0-6).
- "The Escalation Trap" series (3 parts), Medium (medium.com/p/c1976c2f6950).
- ORCID: 0009-0004-9496-6339. STEM competition judge (UC Davis C-STEM, Sacramento STEM Faire) and volunteer AI instructor for high school students. Credentials: Cloud Computing, Stanford Continuing Studies.

EDUCATION

Master of Science (MS), Software Engineering | BITS Pilani

Bachelor of Engineering (BE), Computer Science & Engineering | University of Mysore